

**BANARAS HINDU UNIVERSITY
DEPARTMENT OF CHEMISTRY**

B.Sc. — Semester IV (NEP) Mid Semester Examination, 2025-26

**PHYSICAL CHEMISTRY
Paper Code: CHEMN41 (Mid-Semester)**

Time: 1 Hour

Maximum Marks: 20

*Instructions: Attempt all questions. The figures in the right-hand margin indicate marks.***Question 1.**

- (a) Write the expression of chemical potential of a gas in real gas mixture, solvent and solute in non-ideal liquid solution. [2]
- (b) State physical significance of chemical potential. [1]
- (c) State the relation between fugacity of gas with its pressure at (i) Boyle temperature (ii) very high pressure (iii) very low pressure. [2]
- (d) Derive the Gibbs-Duhem equation and its significance. [3]
- (e) State Raoult's and Henry's law. On what type of solution, each of the law is applicable. [2]
- (f) Why colligative properties of a solution are only dependent on the number of solute particles present not on their chemical identity? [1]
- (g) Using a chemical potential vs temperature plot, show that the boiling point of a liquid increases in solution compared to that of the pure liquid. [2]
- (h) Define Curie and Becquerel units of radioactivity. [2]
- (i) The activity of radioactive isotope reduced by 25 percent after 100 minutes. Calculate the decay constant and half-life of the isotope. [2]
- (j) Using the Bethe notation for the nuclear reactions, complete the following reactions: [2]
 - (i) $^{197}\text{Au}(n, \gamma) \longrightarrow$
 - (ii) $^{27}\text{Al}(\gamma, n) \longrightarrow$

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